

Joe Hollander

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EDUCATION

Columbia College, Columbia University

B.A. Mathematics

New York, NY

Expected 2030

Relevant Coursework: Honors Mathematics

South Burlington High School

South Burlington, VT

GPA: 3.99/4.0 SAT: 1580 PSAT: 1520/1520

2022–2026

Relevant Coursework: AP Calculus BC (5); AP Statistics (5); Calculus III (Multivariable) and Applied Linear Algebra, taken through dual enrollment at the University of Vermont

AWARDS & COMPETITIONS

- **Vermont State Math League Champion**, four consecutive years — undefeated as team captain in 2026
- **2nd place individual**, Vermont State Math League; **1st place**, Advanced Math division
- **1st place statewide**, University of Vermont High School Prize Examination
- **ARML invitee** — 1/16 selected for Vermont's team

RESEARCH

Gold Futures and Leveraged-ETF Replication — independent research paper: [rough draft](#) •

- Tested whether futures portfolios track spot gold and leveraged ETFs better than the ETFs track their own benchmarks
- Built continuous futures series from raw CME data: decoded ticker conventions, implemented two roll methodologies, back-adjusted for splicing, and validated against an independent reference
- Solved optimal replicating portfolios as a **constrained quadratic program** (`cvxpy`); several beat **GLD's own out-of-sample tracking error**
- Found **statistically significant** leverage deviations in **2 of 4** ETFs via t-tests on regression slope and daily tracking bias
- Produced draft of findings using $\text{\LaTeX}$ and `matplotlib`, including a variety of visualizations and tables

PROJECTS

Options Volatility and Flow Analysis

- Replicated the **CBOE VIX methodology** end-to-end on AAPL options, including put-call parity forward extraction and cubic-spline interpolation of Treasury rates
- Priced options via **Black-Scholes** and **Cox-Ross-Rubinstein** and **Leisen-Reimer** trees; solved implied volatility with Newton-Raphson and analytic vega; built an options-flow signal by moneyness and expiry

Systematic Strategy Backtesting

- Backtested mean-reversion and volatility-breakout equity strategies on **NautilusTrader**, an event-driven engine
- Built custom serializable data types and a producer/consumer architecture separating signal from execution; handled warm-up periods and notional-capped sizing

Real-Time Market Data Service

- Containerized a **FastAPI** service streaming multi-exchange order-book data into **ArcticDB**, a versioned time-series database
- Computed a **cost-based VWAP** walking book levels with partial-fill interpolation to estimate execution price at a target notional

Machine Learning Competitions

- **Top 25% of 3,000+** entrants, Kaggle Podcast Listening Time; also competed in the National HS Data Challenge
- Applied cyclical encoding, validated model-based imputation, and **leakage-safe target encoding** via nested cross-validation

SKILLS

- **Languages & Tools:** Python 3, $\text{\LaTeX}$, Git, Docker, `asyncio`, `ccxt`, Databento
- **Scientific Computing & ML:** NumPy, pandas, SciPy, statsmodels, `cvxpy`, Matplotlib, scikit-learn, XGBoost, LightGBM

ACTIVITIES

- Math League (Captain 2026); Math Tutoring; Ping Pong Club President (2024-2026); Varsity Ultimate Frisbee; Varsity and Club Soccer

Soccer Referee — Vermont Soccer Association

- **600+ hours** officiating, center refereeing through the **U19** level; selected as a **State Cup referee**
- Demands split-second judgment under pressure, sustained focus, and conviction in a call under scrutiny